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1-C:

Total Utility: An indifference curve (like in the diagram) represents all combinations of two goods (X and Y) that provide a consumer with the same level of total satisfaction or utility. Since both points M and N lie on the same indifference curve, the total utility remains unchanged when the consumer moves from M to N.

Marginal Utility: The shape of the indifference curve is convex to the origin. As the consumer moves from point M to point N:

- The quantity of good X consumed decreases, and the quantity of good Y consumed increases.
- Due to the law of diminishing marginal utility, as more of a good is consumed, the additional satisfaction (marginal utility) gained from each extra unit decreases.

Therefore, the marginal utility from good X increases (as less is consumed), and the marginal utility from good Y decreases (as more is consumed).

2-B:

Demerit Goods: By definition, demerit goods are goods that are considered to be harmful to the consumer or society but are over-consumed due to information failure (consumers don't realize the long-term harm) or negative externalities.

Since the question asks for a cause of under-consumption, the good should not be a demerit good (it would more likely be a merit good).

Conclusion: This condition should be "no"

External Benefits in Consumption (Positive Externalities): When a good has external benefits, the Marginal Social Benefit (MSB) is greater than the Marginal Private Benefit (MPB). In a free market, consumers only consider their private benefits. Because they ignore the positive impact on others, the good is under-consumed relative to the socially optimum level.

Conclusion: This condition must be "yes" for under-consumption.

3-B:

The lack of clearly defined property rights often leads to the "tragedy of the commons," where resources that are not owned by anyone (common-pool resources) are overused and depleted.

1) When a resource has no owner, no one can charge a price for its use.

2) To the individual user, the marginal private cost of consuming the resource is zero, even though there is a social cost (the cost to society from the resource being depleted).

3) Because the private cost is zero, users consume the resource until their private benefit reaches zero, leading to overconsumption and depletion of the scarce resource.

Options A, C, and D are incorrect as the core issue is the lack of a price signal to reflect scarcity, leading to overuse, not under-allocation or selling at the wrong price by an owner.

4-B:

A cost-benefit analysis for a large public project like a hydroelectric power station is conducted from society's perspective, not an individual or private company's perspective. The goal is to determine if the project benefits society as a whole.

Why (B) is correct: A project should be approved if the overall benefits to society (social benefits) outweigh the overall costs to society (social costs). Social costs include both private costs (e.g., construction costs) and external costs (e.g. environmental damage), and social benefits include both private benefits (e.g., electricity sales) and external benefits (e.g., flood control).

Why (A) is incorrect: "Private benefits are greater than private costs" only considers the profitability for the private entity building the station. It ignores external factors (externalities), which are significant in such projects. A project might be privately profitable but harmful to society overall if external costs are very high.

Why (C) is incorrect: "Social benefits are greater than external costs" is an incomplete comparison. It compares total benefits to only a portion of the total costs (external costs). It ignores the private costs.

Why (D) is incorrect: "External costs are greater than external benefits" suggests that the negative externalities outweigh the positive externalities. This condition, if true, would make the scheme less likely to be approved, as it indicates a net negative impact on third parties. The condition for approval requires total social benefits to exceed total social costs.

5-B:

Price leadership is a characteristic feature of an oligopoly market structure. In an oligopoly, there are only a few dominant firms, and the actions of one firm significantly impact the others. To avoid price wars, firms may tacitly agree to follow the pricing decisions of a single, dominant firm, which acts as the price leader.

6-B:

An oligopoly is a market structure dominated by a small number of large firms. A common measure of market concentration is the concentration ratio, which sums the market shares of the largest firms. A four-firm concentration ratio above 90% typically indicates a strong oligopoly.

For Industry X, the top five firms hold a combined market share of $32\% + 26\% + 22\% + 10\% + 5\% = 95\%$. The top four firms alone hold 90% of the market. This high concentration ratio strongly suggests that Industry X is an oligopoly.

For comparison, in Industry Z, the top five firms hold $17\% + 15\% + 14\% + 12\% + 6\% = 64\%$ of the market. This is lower than Industry X, suggesting Industry Z is more competitive than Industry X (making option D incorrect). Option A is incorrect because high concentration usually suggests low contestability. Option C cannot be determined from market share percentages alone, as output levels are not provided.

7-B:

The diagram shows a negative production externality, as the Marginal Social Cost (MSC) is greater than the Marginal Private Cost (MPC). The free market equilibrium is at point R, where MPC equals the Marginal Private Benefit (MPB) which is also the Marginal Social Benefit (MSB). The socially optimal level of output is at point H, where MSC equals MSB, which corresponds to the output level S.

To move the market from the inefficient output level W (point R) to the socially optimal output level S (point H), the government should implement a policy that increases the private cost of production, internalizing the externality. A specific tax equal to the vertical distance between the MSC and MPC at the optimal output (distance HM) would achieve this goal. This shifts the MPC curve upwards to align with the MSC curve, leading to the socially optimal outcome.

8-C:

Nudge theory is a behavioral approach that suggests indirect suggestions and positive reinforcement can influence people's decisions and behaviors in a predictable way. It avoids direct compulsion, mandates, or significant financial incentives/penalties.

(A) Compelling a tax deduction is a mandate, not a nudge.

(B) Making random inspections acts as a deterrent through fear of penalty, which is not a nudge.

(C) Providing information about tax spending is a form of transparency and social persuasion, which can subtly influence citizens' behavior by appealing to their sense of civic duty and trust, making it an example of a nudge.

(D) Using penalties like fines and imprisonment are forms of direct enforcement and punishment, not a subtle nudge.

9-C:

A lower Gini coefficient indicates a more equal distribution of income. The Gini coefficient is a measure of statistical dispersion intended to represent the income inequality or the wealth inequality within a nation or a social group. A coefficient of 0 expresses perfect equality (everyone has the same income), while a coefficient of 1 (or 100%) expresses maximal inequality (one person has all the income or wealth, and all others have none). Therefore, a lower Gini coefficient means that the income distribution is closer to perfect equality.

A. A faster rate of economic growth

Economic growth refers to the increase in the production of economic goods and services in an economy. A faster rate of economic growth does not necessarily indicate a more equal distribution of income. In fact, rapid economic growth can sometimes lead to an increase in income inequality if the benefits are not distributed evenly across the population.

B. A higher Human Development Index

The Human Development Index (HDI) is a composite statistic of life expectancy, education, and per capita income indicators, which are used to rank countries into four tiers of human development. While a higher HDI generally indicates better overall well-being, it is a measure of development, not specifically income distribution. A country can have a high HDI but still have significant income inequality.

D. A lower tax/GDP ratio

The tax-to-GDP ratio is a measure of a nation's tax revenue relative to its Gross Domestic Product (GDP). A lower ratio means the government collects less tax relative to the size of the economy. Lower tax collection might limit a government's ability to implement redistributive policies (such as social welfare programs or progressive taxation), which are

typically used to achieve a more equal distribution of income. Therefore, a lower ratio is less likely to indicate more equality.

10-C:

A ban on substitutes for cigarettes would likely increase the negative externalities associated with smoking. Negative externalities refer to the costs incurred by third parties not directly involved in the transaction (e.g., secondhand smoke, healthcare costs).

Bans on advertising (A), smoking in public (B), or sales (D) are all policies designed to reduce cigarette consumption or the impact of smoking, thus decreasing negative externalities.

A ban on substitutes, however, would make alternatives like vaping or nicotine patches less accessible, potentially leading more people to continue smoking traditional cigarettes. An increase in the number of people smoking cigarettes would increase the overall negative externalities.

11-B:

The amount by which employment is reduced when a minimum wage is set above the equilibrium wage is determined by the elasticity of demand for labour, which is represented by the marginal revenue product (MRP) curve in a firm's hiring decision. The MRP curve is the demand curve for labour. The steeper (more inelastic) the MRP curve, the smaller the reduction in employment; the flatter (more elastic) the MRP curve, the larger the reduction in employment. The mobility of labour (C) and the wage rate in other industries (D) primarily affect the supply of labour, not the demand for it or the magnitude of the employment reduction directly. The average revenue product curve (A) is not the relevant curve for determining the optimal level of labour demand.

12-B:

A trade union's power to increase wages without significantly reducing employment is maximized when the demand for the labor it supplies is inelastic.

(B) The economy is experiencing rising employment. In a period of rising employment, the demand for the industry's goods and thus the demand for labor is high. Firms are expanding and need workers, which increases the bargaining power of the union and allows for wage increases with minimal impact on employment levels.

(A) The economy is experiencing a fall in the price of capital. A fall in the price of capital makes it a cheaper substitute for labor. This would likely encourage firms to substitute capital for labor if wages rise, thereby reducing the union's power and leading to job losses if wages increase.

(C) The price elasticity of demand for the industry's goods is equal to 1. While better than elastic demand, the ideal condition for union power is inelastic demand (less than 1). Unit elastic demand means total revenue remains the same if prices change, but it doesn't prevent a reduction in employment if firms decide to cut output in response to higher costs.

(D) The price elasticity of demand for the industry's goods is greater than 1. Elastic demand for the product means that if the firm raises prices (due to higher wages), the quantity demanded will fall more than proportionally, leading to a significant drop in production and a large reduction in employment. This condition minimizes union power to raise wages without job losses.

13-D:

In the given consumption equation, $C=a+by$

C is the total consumption, Y is the national income, a is the autonomous consumption (consumption that occurs even with zero income), and b is the induced consumption, which represents the marginal propensity to consume (MPC). The MPC is the change in consumption caused by a one-unit change in income. It is assumed to be between zero and one.

14-A:

Autonomous expenditure is the level of aggregate demand when income is zero, represented by the vertical intercept of the AMD1 curve, which is point O on the y-axis.

Induced expenditure is the portion of expenditure that varies with income. The total aggregate demand (AMD) at any income level is the sum of autonomous and induced expenditure. The difference between total AMD and the autonomous component (O) at a given income level represents the induced expenditure at that level.

At income level O (zero income), total AMD is autonomous expenditure (O), and induced expenditure is zero. The difference is O.

As income increases (moving from O to B to C to D), induced expenditure increases (the vertical distance between the x-axis and the AMD1 curve increases).

The difference between autonomous and induced expenditure is greatest at the lowest level of income shown, which is A, where induced expenditure is minimal (zero) and autonomous expenditure is at its maximum relative to induced expenditure. The absolute difference is the full value of the autonomous expenditure.

15-D:

The accelerator theory states that investment is determined by the rate of change of national income (or output). Investment will be lower than in the previous year if the increase in national income is smaller than the previous year's increase, or if national income is falling.

(A) a lower marginal propensity to consume than in the previous year: The marginal propensity to consume (MPC) relates to the Keynesian multiplier effect, not directly to the accelerator theory of investment. A lower MPC would generally lead to lower overall demand, but the accelerator theory focuses specifically on the change in national income as the driver of investment.

(B) a lower marginal propensity to save than in the previous year: The marginal propensity to save (MPS) is related to the MPC ($MPC + MPS = 1$). This is also related to the multiplier and not the direct cause of investment changes in the accelerator model.

(C) a smaller fall in national income than occurred in the previous year: If national income is falling, investment is likely to be negative or zero (disinvestment). A smaller fall (e.g., -5% vs

-10% the previous year) actually represents an improvement in the change in national income, which would cause investment to be higher than the previous year (less negative).

(D) a smaller rise in national income than occurred in the previous year: This is the correct answer. If national income was rising last year (e.g., +10%) but is rising less this year (e.g., +5%), the change in the rate of growth is negative. According to the accelerator principle, a deceleration in the growth of national income leads to a reduction in net investment compared to the previous period.

16-B:

Sustainable economic development focuses on meeting the needs of the present without compromising the ability of future generations to meet their own needs. Option B accurately captures this balance, as it emphasizes both the improvement of living standards (economic development) and the preservation of the environment (sustainability).

Options A and C describe general economic growth (increase in actual output or productive potential) but do not inherently include the crucial environmental or social components of sustainability.

Option D focuses solely on living standards without mentioning the environmental aspect.

17-B:

The natural rate of unemployment is the long-run equilibrium unemployment rate, determined by structural and frictional factors.

(A) A decrease in information about vacancies: This makes it harder for job seekers to find available jobs, increasing frictional unemployment and thus the natural rate.

(B) A decrease in trade union power: Stronger unions can negotiate higher wages, which may lead to some job losses and higher structural unemployment. Therefore, a decrease in trade union power is likely to decrease the natural rate of unemployment, making it the least likely option to increase it.

(C) An increase in unemployment benefits: More generous benefits reduce the urgency for the unemployed to find work quickly, increasing the duration of unemployment and the natural rate.

(D) An increase in immigration: While immigration can increase the labor supply, a large influx of workers in the short term, especially if they lack specific skills or face language barriers, can increase frictional and structural unemployment as they search for jobs and integrate into the labor market.

18-D:

In Keynesian economic theory, specifically the Liquidity Preference Theory, interest is viewed as the price paid for the use of money (loanable funds). It is the reward for parting with liquidity (cash) for a specified period. Keynes argued that people have a preference for holding liquid assets (cash) for transactional, precautionary, and speculative motives. The interest rate must be high enough to compensate individuals for giving up this convenience and safety of holding cash. The other options relate to different economic concepts or theories.

(A) to bring national income and expenditure into line: In the Keynesian model, the equality between national income and aggregate expenditure is achieved through adjustments in the level of output and employment, via the multiplier effect, not primarily through the rate of interest.

(B) to control the level of economic growth: While monetary policy (which involves manipulating interest rates) is used to influence economic growth, this describes a policy application of interest rates, not their fundamental purpose or determination within the core Keynesian theory of interest itself.

(C) to create a balance between savings and investment: This is the classical economic view of interest. In Keynesian theory, savings and investment are brought into equilibrium by changes in the level of national income, not the interest rate.

19-A:

The quantity theory of money is represented by the equation $MV=PT$ or $MV=PY$

The variables in the equation are defined as:

M is the money supply in circulation within the economy. This typically includes more than just physical notes and coins (e.g., bank deposits).

V is the velocity of money, which is the average number of times a unit of money is spent during a given time period. P is the average price level of goods and services in the economy.

T stands for the total volume of real transactions or real output (real GDP) during the year.

Statement A is not correct because M generally refers to a broader measure of money supply (e.g., $M1$, $M2$), not just the total value of notes and coins in circulation. The other statements (B, C, and D) accurately describe their respective variables in the context of the equation.

20-B:

Export subsidies make a country's goods cheaper for foreign consumers, which increases demand for exports and helps to solve the trade deficit. The increased production for export also leads to higher employment, addressing the high unemployment problem. While increased demand can put upward pressure on prices (inflation), this policy is often presented as a potential solution for all three issues in some economic models, particularly when compared to other options:

Devaluation of the currency (A) also helps the trade deficit and unemployment but directly increases the domestic price of imports, significantly contributing to inflation.

Lower interest rates (C) boost domestic demand and employment but typically worsen both inflation and the trade deficit (as imports become more attractive and capital flows out).

Tariffs on imports (D) help the trade deficit and employment in import-competing industries, but they increase consumer prices (inflation) and risk retaliatory trade measures from other countries.

Therefore, within the context of a multiple-choice question seeking one "most likely" solution to all three problems, export subsidies (B) might be considered the intended answer, assuming the inflationary impact is manageable or less severe than devaluation.

21-A:

Quantitative easing (QE) is a monetary policy where a central bank buys government bonds or other financial assets to inject money into the economy and increase the money supply.

Internal Value: An increase in the money supply typically leads to an increase in the general price level (inflation) in the domestic economy. As prices rise, the purchasing power of each unit of currency decreases, meaning the internal value of the US dollar decreases.

External Value: The increased supply of the US dollar on the foreign exchange market tends to depress its value relative to other currencies. This depreciation (decrease in external value) makes US exports cheaper and imports more expensive.

22-B:

A balance of payments deficit with a floating exchange rate causes the currency to depreciate. This depreciation makes exports cheaper and imports more expensive.

Price elasticity of demand for exports: The demand should be elastic. This means that the fall in the price of exports (due to depreciation) leads to a proportionally larger increase in the quantity of exports sold. This significantly increases aggregate demand in the economy.

Unemployment rate: The unemployment rate should be low. A low unemployment rate indicates that the economy is operating near full capacity. When aggregate demand increases significantly in a near-full capacity economy, firms cannot easily increase output, leading to a general increase in the price level (demand-pull inflation).

Therefore, PED for exports should be elastic and the unemployment rate should be lower to create the most demand-pull inflationary pressure in an economy with a floating exchange rate.

23-B:

The Phillips curve analysis suggests an inverse relationship between unemployment and inflation; as one decreases, the other tends to increase, and vice versa.

Year 1 to Year 2: Both unemployment and inflation decreased, which is not consistent with the Phillips curve relationship. (see the table)

Year 2 to Year 3: Unemployment decreased (from 6.2% to 5.8%) while inflation increased (from 1.6% to 1.8%). This inverse relationship is consistent with the Phillips curve analysis.

Year 3 to Year 4: Both unemployment and inflation increased, which is not consistent with the Phillips curve relationship. (see the table)

Year 4 to Year 5: Both unemployment and inflation stayed the same, which is not consistent with the Phillips curve relationship. (see the table)

24-D:

Before the tariffs were removed, the price was $PW+T$. At this price, the domestic supply was $Q2$ and the domestic demand was $Q3$. Therefore, the level of imports was the difference between domestic demand and supply, which is the quantity $Q3-Q2$ represented by the segment.

After the tariffs were removed, the price fell to the world price PW . At this new price, the domestic supply was $Q1$ and the domestic demand was $Q4$. The new level of imports was $Q4-Q1$ represented by the segment

The change in imports resulting from the removal of tariffs is the new level of imports minus the old level of imports:

$$\text{Change in imports} = (Q4 - Q1) - (Q3 - Q2)$$

This corresponds to option D, expressed as $Q1Q4 - Q2Q3$ in the option's notation.

25-B:

A current account deficit means a country is importing more goods, services, income, and transfers than it is exporting. To fix this, the country needs a long-term solution that brings in more foreign currency or reduces the outflow.

A is incorrect because increasing borrowing only provides short-term relief and increases future debt obligations.

B is a strong long-term solution. Foreign direct investment (FDI) brings in foreign currency and often leads to increased exports and reduced imports in the long run, directly addressing the deficit's root causes.

C is incorrect because revaluing the currency (making it stronger) makes exports more expensive and imports cheaper, which would likely worsen the deficit.

D is incorrect because reducing foreign exchange reserves is a short-term measure to prop up the currency, not a long-term solution to the underlying deficit problem.

26-B:

An expenditure-reducing policy aims to reduce aggregate demand and national income to decrease the demand for imports. This policy is generally more successful than an expenditure-switching policy (like devaluation) when the Marshall-Lerner condition is not met, meaning the sum of the price elasticities of demand for exports and imports is less than 1 ($X+M < 1$)

Under these conditions, devaluation is ineffective or worsens the current account.

In options A, B, and C, the sum of elasticities is 0.4, 0.4, and 0.8 respectively, all less than 1. In option D, the sum is 1.2, so switching would be effective.

Between options A, B, and C, the effectiveness of an expenditure-reducing policy also depends on the marginal propensity to import (MPM), which is the fraction of any change in income spent on imports. A higher MPM means that a given reduction in national income leads to a greater reduction in imports, making the policy more effective.

Option B has the highest MPM (0.4) among the options where the Marshall-Lerner condition is not met, making the expenditure-reducing policy the most successful relative to an expenditure-switching policy in this scenario.

27-A:

The J-curve effect describes the phenomenon where a country's trade balance initially worsens after a currency depreciation (short run) and then improves (long run). This occurs when the Marshall-Lerner condition, which states that the sum of the price elasticities of demand for exports and imports must be greater than 1 for depreciation to improve the trade balance, is not met in the short run but is met in the long run.

For the short run (worsening current account), the sum of short-run elasticities must be less than 1: $(X+M < 1)$

For the long run (improving current account), the sum of long-run elasticities must be greater than 1: $(X+M > 1)$

Option A satisfies both the SR and LR conditions of a J-curve effect.

Option A: $0.5+0.2=0.7 < 1$ (short run condition met). Long run: $0.6+0.6=1.2 > 1$ (long run condition met).

* Option B: $0.5+0.6=1.1 > 1$ (short run condition not met).

* Option C: $0.6+0.6=1.2 > 1$ (short run condition not met).

* Option D: $1.2+1.0=2.2 > 1$ (short run condition not met).

Therefore, option A is the correct answer.

28-A:

Low-income countries are typically characterized by low average incomes, limited industrialization, and high levels of poverty. These conditions often lead to a high death rate and low levels of productivity.

- High death rates are often linked to inadequate healthcare infrastructure and sanitation.
- Low productivity levels result from limited access to technology, education, and capital.
- High gross domestic product (GDP) and low infant mortality are characteristics of high-income countries.

- Low birth rates are usually associated with high-income countries, while a large proportion of income from primary industries (like agriculture) is typical of low-income countries.
- A large tertiary sector (services) is characteristic of high-income countries

Therefore option A meets both the characteristics of a low income country.

29-B:

A depreciation of a currency means that its value has fallen relative to other currencies. This makes exports cheaper for foreigners and imports more expensive for domestic consumers, assuming demand is relatively elastic (Marshall-Lerner condition).

Increasing the rate of growth: Cheaper exports boost export-oriented industries, increasing aggregate demand and stimulating economic growth. Therefore, depreciation is likely to increase the rate of growth.

Reducing the current account deficit: The increase in exports and decrease in imports improves the balance of trade, which is the main component of the current account. This helps to reduce the current account deficit.

Reducing the rate of unemployment: The increase in demand for domestically produced goods and services (both from exports and import substitution) encourages firms to hire more workers, thus reducing unemployment.

Option B correctly indicates "yes" for all three effects.

30-B:

A- When a country receives foreign aid, foreign currency (like dollars or euros) enters the country. This increases the supply of foreign currency and it can stabilize or strengthen the exchange rate. So the exchange rate could improve.

B- Financial account of the balance of payments (least likely to improve)

The financial account records things like:

- Foreign direct investment (FDI)
- Portfolio investment
- Loans and financial assets

Foreign aid is usually recorded as a transfer in the current account, not in the financial account. So increasing aid does not directly improve the financial account. Therefore this is least likely to improve.

C- Aid is often used as a tool to foster goodwill and strengthen diplomatic and political ties between nations.

D- The influx of foreign aid directly increases the central bank's holdings of foreign currency reserves

Therefore option B is correct because foreign aid does not directly affect the financial account of the balance of payments.